Investment Portfolio Analysis & Construction

Composition & Market Outlook

Ticker	Name	Composition	Market Outlook
VOO	Vanguard S&P 500	Tracks the 500 largest US companies by market cap. Top holdings: Apple, Microsoft, Nvidia, Amazon, Meta. ~30% in tech, rest spread across healthcare, financials, consumer.	US large-caps benefit from resilient corporate earnings and the start of the Fed rate-cutting cycle (first cut Sep 2024). AI-driven capex from mega-caps supports continued growth.
URTH	iShares MSCI World	Tracks large/mid-cap stocks across 23 developed markets. ~70% US, rest in Japan, UK, Canada, Europe. Effectively a more diversified version of VOO.	Global developed-market recovery underway as inflation cools across the US and Europe. Weaker USD from rate cuts could lift non-US holdings when translated back.
EWM	iShares MSCI Malaysia	Broad exposure to Malaysian equities — banks (Maybank, CIMB), plantations (Sime Darby), telecoms, utilities. Heavily financials-weighted.	Malaysia benefits from China+1 supply chain diversification — FDI into Penang semiconductor corridor. Ringgit strengthening as Fed cuts narrow the rate differential.
EWJ	iShares MSCI Japan	Large/mid-cap Japanese equities — Toyota, Sony, Mitsubishi, Keyence. Mix of autos, industrials, tech, financials.	BOJ ended negative rates in March 2024 — signals confidence in the economy. Corporate governance reforms (Tokyo Stock Exchange push for higher ROE) driving re-rating of Japanese equities. Weak yen boosts exporters.
GLD	SPDR Gold Shares	Tracks the spot price of gold bullion. No equity exposure — purely a commodity play.	Gold hit record highs in 2024, driven by central bank buying (China, India), geopolitical hedging (Ukraine, Middle East), and anticipated Fed rate cuts which lower the opportunity cost of holding gold.
	Invesco	Tracks the Nasdaq-100: the 100 largest non-financial Nasdaq-listed companies.	AI infrastructure buildout is the dominant theme — Nvidia data center revenue tripled YoY.

QQQ	QQQ Trust	~60% in mega-cap tech (Apple, Nvidia, Microsoft, Broadcom, Meta).	Hyperscaler capex from Microsoft, Google, Amazon flowing to semiconductor and cloud names in the index.
IWM	iShares Russell 2000	Tracks the 2,000 smallest stocks in the Russell 3000. Broad sector exposure — industrials, healthcare, financials, tech. More domestically oriented than large-caps.	Small-caps historically outperform early in rate-cutting cycles as cheaper borrowing lifts smaller, more leveraged companies. Valuations are at multi-decade discounts relative to large-caps.
MCHI	iShares MSCI China	Large/mid-cap Chinese equities — Tencent, Alibaba, Meituan, PDD, JD.com. Heavy internet/consumer discretionary exposure.	Sentiment remains challenged — property sector deleveraging, weak consumer confidence, and US-China tensions weigh on valuations. However, stocks trade at deep discounts (MSCI China P/E ~10x vs S&P 500 ~21x), creating a contrarian case if stimulus materialises.

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```
import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt
import statsmodels.api as sm
from scipy.optimize import minimize
import os

# Output directory
OUT_DIR = os.path.join(os.getcwd(), 'out')
os.makedirs(OUT_DIR, exist_ok=True)



tickers = ["VOO", "URTH", "EWM", "EWJ", "GLD", "QQQ", "IWM", "MCHI"
names = {
    'VOO': 'Vanguard S&P 500', 'URTH': 'iShares MSCI World',
    'EWM': 'iShares MSCI Malaysia', 'EWJ': 'iShares MSCI Japan',
    'GLD': 'SPDR Gold Shares', 'QQQ': 'Invesco QQQ Trust',
    'IWM': 'iShares Russell 2000', 'MCHI': 'iShares MSCI China'
}

# Assume the risk-free rate is 4%
rf = 0.04

data = yf.download(tickers, start="2019-01-01", end="2024-12-31", a
data = data.dropna()

data.tail()
```

```
[*********************100%***********************]  8 of 8 completed
```

Out[171…

Ticker	EWJ	EWM	GLD	IWM	MCHI	Q(
Date						
2024-12-23	63.577255	22.781874	240.960007	218.558975	46.261768	520.2315
2024-12-24	63.539005	23.013161	241.440002	220.513474	46.672638	527.2858
2024-12-26	64.275276	23.302273	243.070007	222.773972	46.790031	526.9275
2024-12-27	64.762939	23.427553	241.399994	219.526352	46.437859	519.9231
2024-12-30	64.217911	23.639568	240.630005	217.838379	45.929165	513.0081

1. Return & Risk Analysis

Determine each asset's standalone attractiveness using : Return, Risk and Sharpe Ratio. All annualized.

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```
daily_returns = data.pct_change().dropna()

mean = daily_returns.mean()
sd = daily_returns.std()

annual_return = mean * 252
annual_volatility = sd * np.sqrt(252)
sharpe_ratios = (annual_return - rf) / annual_volatility

best_individual = sharpe_ratios.idxmax()
best_sr = sharpe_ratios.max()
print(f"Best individual Sharpe: {best_individual} ({best_sr:.3f}) —

summary = pd.DataFrame({
    "Annualised Return" : annual_return,
    "Annualised Volatility" : annual_volatility,
    "Sharpe Ratio" : sharpe_ratios
})

summary
```

```
Best individual Sharpe: QQQ (0.811) — Section 4 portfolio must beat
this to justify multi-asset construction
```

Out[172…

Ticker	Annualised Return	Annualised Volatility	Sharpe Ratio
EWJ	0.080246	0.179728	0.223929
EWM	0.017993	0.169303	-0.129983
GLD	0.125640	0.149518	0.572773
IWM	0.129129	0.258254	0.345122
MCHI	0.046650	0.303857	0.021884
QQQ	0.237272	0.243364	0.810603
URTH	0.150171	0.192157	0.573340
VOO	0.179666	0.200997	0.694869

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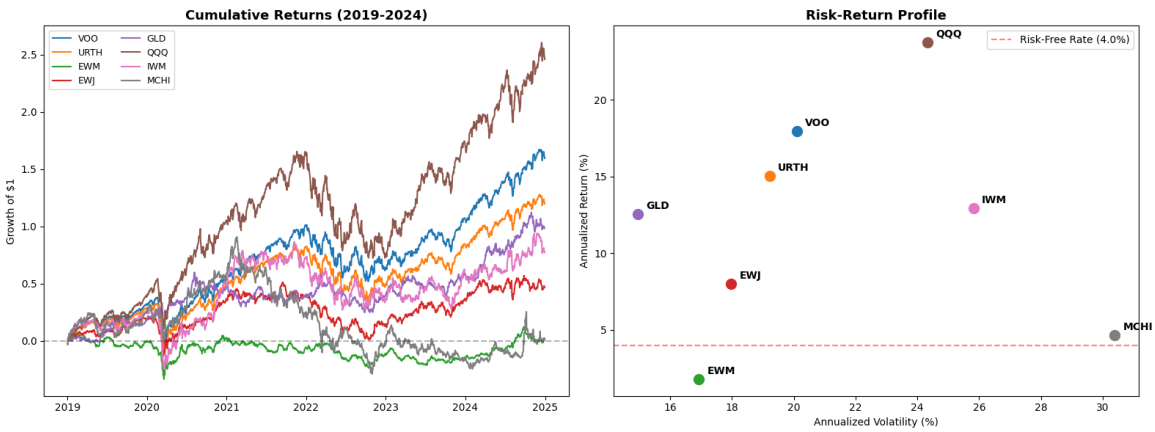
```
cum_returns = (1+daily_returns).cumprod() - 1

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

# Cumulative returns
for t in tickers:
    axes[0].plot(cum_returns.index, cum_returns[t], label=f'{t}', l
axes[0].set_title('Cumulative Returns (2019-2024)', fontweight='bol
axes[0].set_ylabel('Growth of $1')
axes[0].legend(fontsize=9, ncol=2)
axes[0].axhline(y=0, color='black', linestyle='--', alpha=0.3)

# Risk-return scatter
for t in tickers:
    axes[1].scatter(annual_volatility[t]*100, annual_return[t]*100,
    axes[1].annotate(t, (annual_volatility[t]*100, annual_return[t]
                     textcoords="offset points", xytext=(8, 5), fon
axes[1].set_xlabel('Annualized Volatility (%)')
axes[1].set_ylabel('Annualized Return (%)')
axes[1].set_title('Risk-Return Profile', fontweight='bold', fontsiz
axes[1].axhline(y=rf*100, color='red', linestyle='--', alpha=0.5, l
axes[1].legend()

plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, 'risk_return_profile.png'), dpi=1
plt.show()
```



Key Findings:

1. QQQ has the highest return but high volatility — it dominates on Sharpe among individual assets
2. GLD has moderate return with moderate volatility → If GLD's correlation with QQQ is low, it can be a diversifier (confirmed in Section 2)

2. Correlation & Diversification Analysis

Determine if assets can be *combined* to reduce portfolio volatility.

$$\sigma_portfolio^2 = w_1^2\sigma_1^2 + w_2^2\sigma_2^2 + 2 \cdot w_1 \cdot w_2 \cdot \sigma_1 \cdot \sigma_2 \cdot \rho_{12}$$

we look for assets with low correlation (close to 0 or negative)

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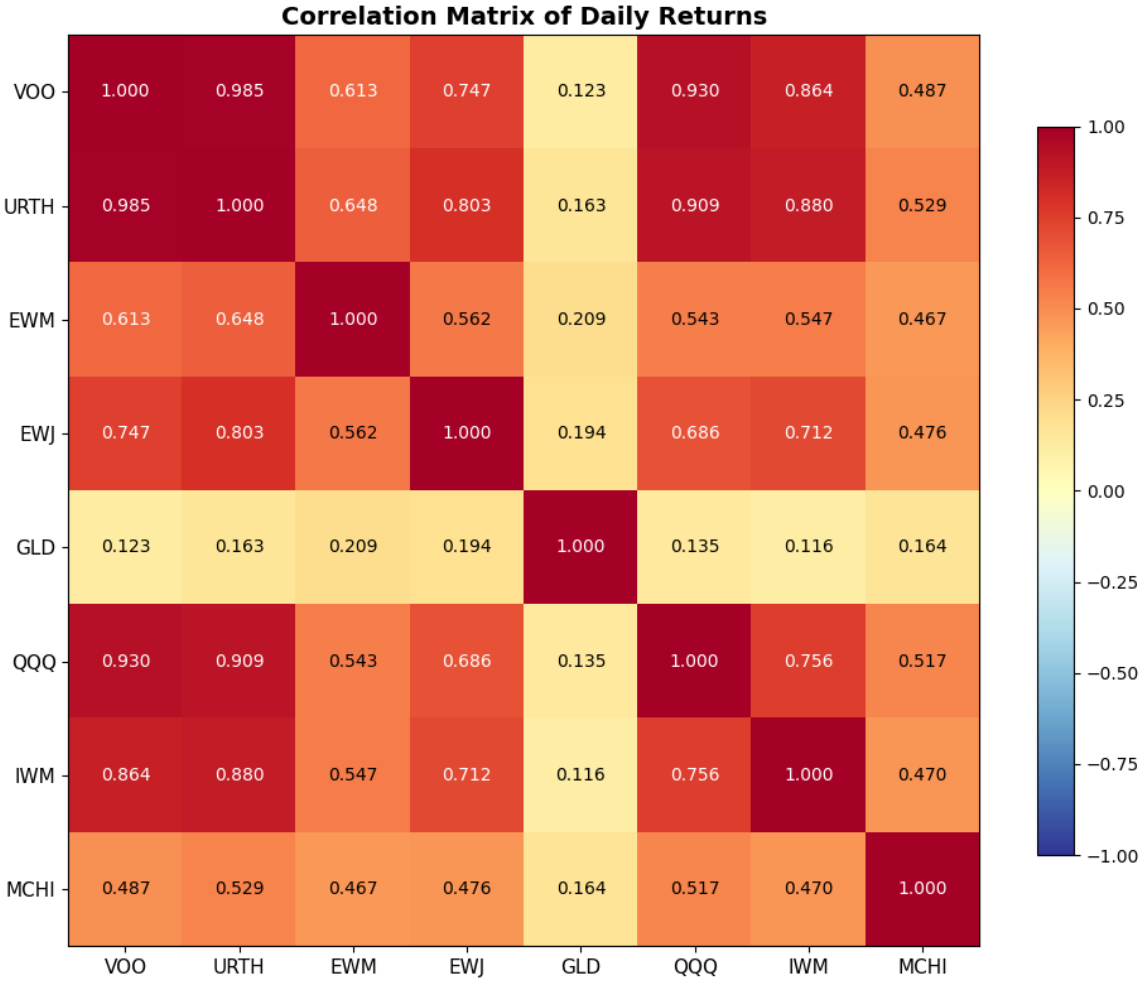
```
# Correlation Matrix
corr_matrix = daily_returns[tickers].corr()

# Draw Correlation Matrix as Heat Map
fig, ax = plt.subplots(figsize=(10, 8))
im = ax.imshow(corr_matrix.values, cmap='RdYlBu_r', vmin=-1, vmax=1

ax.set_xticks(range(len(tickers)))
ax.set_yticks(range(len(tickers)))
ax.set_xticklabels(tickers, fontsize=11)
ax.set_yticklabels(tickers, fontsize=11)

for i in range(len(tickers)):
    for j in range(len(tickers)):
        ax.text(j, i, f'{corr_matrix.values[i, j]:.3f}',
                ha='center', va='center', fontsize=10,
                color='white' if abs(corr_matrix.values[i, j]) > 0.

plt.colorbar(im, ax=ax, shrink=0.8)
ax.set_title('Correlation Matrix of Daily Returns', fontweight='bol
plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, 'correlation_matrix.png'), dpi=15
plt.show()
```

```
print("\nKey Diversification Insights:")
print(f"  - QQQ-GLD correlation: {corr_matrix.loc['QQQ','GLD']:.3f}
print(f"  - Lowest overall: MCHI-GLD ({corr_matrix.loc['MCHI','GLD'
print(f"  - Highest correlation: VOO-URTH ({corr_matrix.loc['VOO','
```




```
Key Diversification Insights:
  - QQQ-GLD correlation: 0.135 → key pair (QQQ is highest-return ass
et from Section 1)
  - Lowest overall: MCHI-GLD (0.164) — but MCHI has negative return
s, low correlation alone isn't useful
  - Highest correlation: VOO-URTH (0.985) → near-redundant
```

Key Findings :

1. GLD has near-zero correlation with QQQ. We confirm that GLD can be used as diversifier for QQQ
2. US equity ETFs (VOO, URTH, QQQ, IWM) are correlated 0.8-0.95. Holding mutliple US equity adds little value to diversification.
3. MCHI has low correlation with most assets but negative returns. Unattractive

3. Beta Estimation via Linear Regression

Decompose each asset's total risk into two components:

- **Systematic risk (B)**
- **Idiosyncratic risk (e)**

The regression model:

`R_asset - Rf = α + β × (R_market - Rf) + ε`

After regression, we compare actual vs required return using CAPM:

`Required Return = Rf + β × (R_market - Rf)`

note that actual return - required return = alpha. Hence, we look for assets that outperforms what the market pays for its level of risk (positive alpha)

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```
# Beta estimation via OLS regression against Market ( VOO as proxy
market_excess = daily_returns['VOO'] - rf/252

betas = {}
alphas = {}
r_squared = {}
capm_expected = {}

fig, axes = plt.subplots(2, 4, figsize=(18, 9))
axes = axes.flatten()

for idx, ticker in enumerate(tickers):

    asset_excess = daily_returns[ticker] - rf/252

    X = sm.add_constant(market_excess)

    model = sm.OLS(asset_excess, X).fit()

    betas[ticker] = model.params.iloc[1]
    alphas[ticker] = model.params.iloc[0] * 252  # Annualize alpha
    r_squared[ticker] = model.rsquared

    # CAPM expected return: E(Ri) = Rf + Beta * (E(Rm) - Rf)
    market_premium = annual_return['VOO'] - rf
    capm_expected[ticker] = rf + betas[ticker] * market_premium

    # Scatter plot
    ax = axes[idx]
    ax.scatter(market_excess.values, asset_excess.values, alpha=0.1
    x_line = np.linspace(market_excess.min(), market_excess.max(),
    y_line = model.params.iloc[0] + model.params.iloc[1] * x_line
    ax.plot(x_line, y_line, 'r-', linewidth=2,
            label=f'β={betas[ticker]:.2f}, R²={r_squared[ticker]:.2
    ax.set_title(f'{ticker}', fontweight='bold')
    ax.legend(fontsize=9)
    ax.set_xlabel('Market Excess Return')
    ax.set_ylabel('Asset Excess Return')

plt.suptitle('Beta Estimation: OLS Regression of Excess Returns vs
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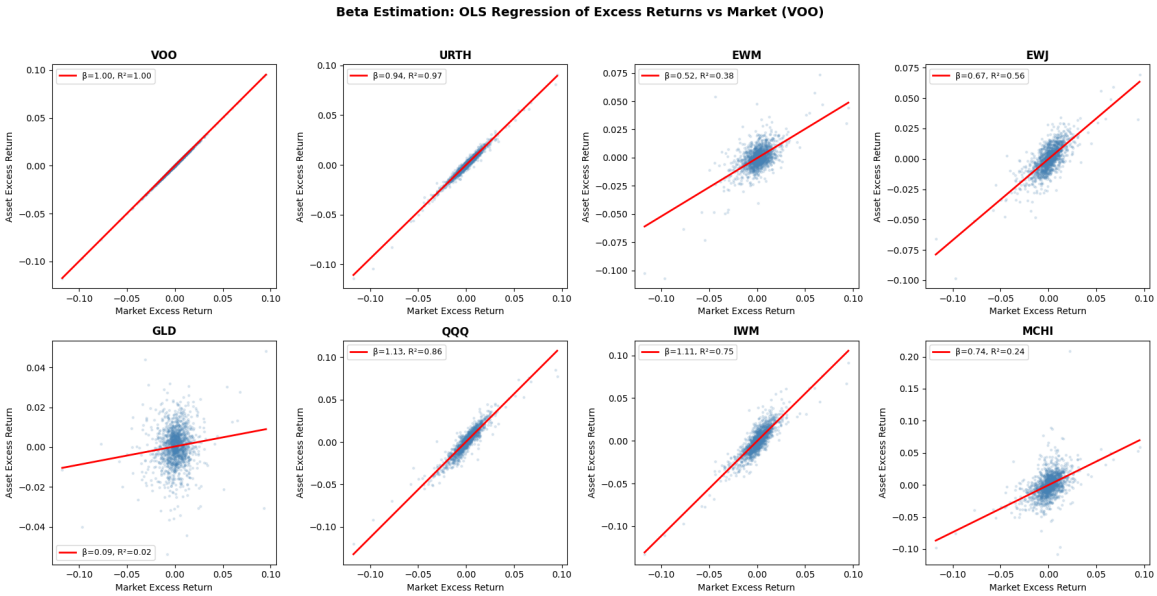
```
                fontweight='bold', fontsize=14, y=1.02)
plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, 'beta_regression.png'), dpi=150,
plt.show()

betas_array = np.array([betas[t] for t in tickers])

# CAPM Summary
capm_df = pd.DataFrame({
    'Beta (b)': [betas[t] for t in tickers],
    'Alpha (a), annualized)': [f"{alphas[t]*100:.2f}%" for t in tic
    'R-squared': [f"{r_squared[t]:.3f}" for t in tickers],
    'CAPM Expected Return (%)': [f"{capm_expected[t]*100:.2f}" for
    'Actual Return (%)': [f"{annual_return[t]*100:.2f}" for t in ti
}, index=tickers)

print("=" * 80)
print("CAPM BETA ESTIMATION RESULTS")
print("=" * 80)
capm_df
```




```
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CAPM BETA ESTIMATION RESULTS
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	Beta (b)	Alpha (a), annualized)	R-squared	CAPM Expected Return (%)	Actual Return (%)
VOO	1.000000	0.00%	1.000	17.97	17.97
URTH	0.942059	-2.14%	0.971	17.16	15.02
EWM	0.516057	-9.41%	0.375	11.21	1.80
EWJ	0.668389	-5.31%	0.559	13.34	8.02
GLD	0.091434	7.29%	0.015	5.28	12.56
QQQ	1.125576	4.01%	0.864	19.72	23.73
IWM	1.109779	-6.59%	0.746	19.50	12.91
MCHI	0.735836	-9.61%	0.237	14.28	4.66

Key Findings:

1. GLD's near-zero beta (0.091) means adding it to a portfolio reduces systematic risk without proportionally reducing return. Assumption of an optimum 2-asset portfolio of QQQ and GLD should hold: GLD provides alpha with no beta cost.

4. Mean-Variance Optimization & Efficient Frontier

Find the portfolio weights that **maximize Sharpe ratio** (Section 1). The optimizer uses the **covariance matrix**, which encodes all the return, volatility, and correlation information (Section 2).

We use Monte Carlo simulation to randomly generate 15,000 portfolios and map the feasible risk-return space. We maximise Sharpe ratio using SLSQP constrained optimisation to return:

1. **Max Sharpe Portfolio** — the weight combination with the best return per unit of total risk
2. **Min Variance Portfolio** — the weight combination with the lowest total volatility (most conservative)
3. **Capital Market Line (CML)** — the line from Rf through the Max Sharpe point. Any portfolio on this line is achievable by mixing the optimal portfolio with cash. Nothing *above* the CML is achievable without leverage.

Conditions:

1. Portfolio Sharpe must beat `best_individual` from section 1 —

otherwise multi-asset construction adds no value

2. The optimizer should select QQQ + GLD — confirming section 2's correlation prediction
3. Portfolio beta (using Section 3's `betas_array`) should be low — confirming GLD dilutes QQQ's beta

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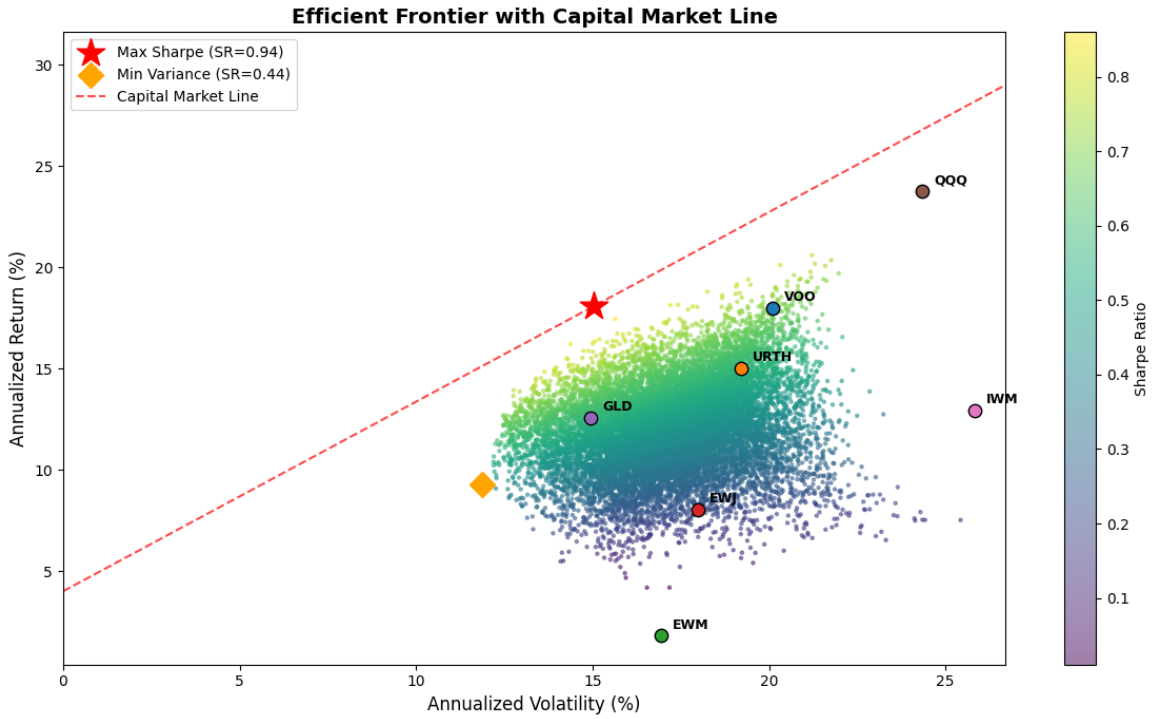
```
# Covariance matrix (annualized)
cov_matrix = daily_returns[tickers].cov() * 252
mean_returns = annual_return[tickers].values

n_assets = len(tickers)

def portfolio_performance(weights, mean_returns, cov_matrix):
    ret = np.dot(weights, mean_returns)
    vol = np.sqrt(np.dot(weights.T, np.dot(cov_matrix.values, weigh
    return ret, vol

def neg_sharpe(weights, mean_returns, cov_matrix, rf):
    ret, vol = portfolio_performance(weights, mean_returns, cov_mat
    return -(ret - rf) / vol

# Constraints: weights sum to 1, each weight between 0 and 1 (long
constraints = ({'type': 'eq', 'fun': lambda w: np.sum(w) - 1})
bounds = tuple((0, 1) for _ in range(n_assets))

# Monte Carlo simulation for efficient frontier visualization
n_portfolios = 15000
results = np.zeros((3, n_portfolios))
weights_record = []

np.random.seed(42)
for i in range(n_portfolios):
    w = np.random.dirichlet(np.ones(n_assets))
    weights_record.append(w)
    ret, vol = portfolio_performance(w, mean_returns, cov_matrix)
    results[0, i] = vol * 100
    results[1, i] = ret * 100
    results[2, i] = (ret - rf) / vol

# Optimal portfolio (max Sharpe)
init_guess = np.array([1/n_assets] * n_assets)
opt_sharpe = minimize(neg_sharpe, init_guess, args=(mean_returns, c
                      method='SLSQP', bounds=bounds, constraints=co
opt_ret, opt_vol = portfolio_performance(opt_sharpe.x, mean_returns

# Minimum variance portfolio
def portfolio_vol(weights, cov_matrix):
    return np.sqrt(np.dot(weights.T, np.dot(cov_matrix.values, weig

min_var = minimize(portfolio_vol, init_guess, args=(cov_matrix,),
                   method='SLSQP', bounds=bounds, constraints=const
mv_ret, mv_vol = portfolio_performance(min_var.x, mean_returns, cov

opt_sr   = (opt_ret - rf) / opt_vol
```

```
opt_beta = np.dot(opt_sharpe.x, betas_array)

print(f"Optimal Sharpe Portfolio: Return={opt_ret*100:.2f}%, Vol={o
print(f"Min Variance Portfolio:   Return={mv_ret*100:.2f}%, Vol={mv

print(f"\nValidation against Section 1-3 predictions:")
top_holdings = pd.Series(opt_sharpe.x, index=tickers).nlargest(2)
print(f"  1. Portfolio Sharpe ({opt_sr:.3f}) vs {best_individual} (
print(f"  2. Top holdings: {', '.join([f'{t} ({w*100:.1f}%)' for t,
print(f"  3. Portfolio beta = {opt_beta:.3f} → crash -10% implies ≈
```

```
Optimal Sharpe Portfolio: Return=18.07%, Vol=15.03%, Sharpe=0.936
Min Variance Portfolio:   Return=9.27%, Vol=11.86%, Sharpe=0.444

Validation against Section 1-3 predictions:
  1. Portfolio Sharpe (0.936) vs QQQ (0.811) → PASS
  2. Top holdings: GLD (50.7%), QQQ (49.3%)
  3. Portfolio beta = 0.601 → crash -10% implies ≈6.0% vs QQQ alone
≈11.3%
```

In [177…

```
# Plot efficient frontier
fig, ax = plt.subplots(figsize=(12, 7))

scatter = ax.scatter(results[0], results[1], c=results[2], cmap='vi
                     marker='o', s=5, alpha=0.5)
plt.colorbar(scatter, ax=ax, label='Sharpe Ratio')

# Optimal portfolio
ax.scatter(opt_vol*100, opt_ret*100, marker='*', color='red', s=400
           label=f'Max Sharpe (SR={((opt_ret-rf)/opt_vol):.2f})')
# Min variance
ax.scatter(mv_vol*100, mv_ret*100, marker='D', color='orange', s=15
           label=f'Min Variance (SR={((mv_ret-rf)/mv_vol):.2f})')

# Individual assets
for t in tickers:
    ax.scatter(annual_volatility[t]*100, annual_return[t]*100, s=80
               edgecolors='black', linewidth=1)
    ax.annotate(t, (annual_volatility[t]*100, annual_return[t]*100)
                textcoords="offset points", xytext=(8, 5), fontsize

# Capital Market Line
cml_x = np.linspace(0, max(results[0])*1.1, 100)
cml_y = rf*100 + ((opt_ret-rf)/opt_vol) * cml_x
ax.plot(cml_x, cml_y, 'r--', linewidth=1.5, alpha=0.7, label='Capit

ax.set_xlabel('Annualized Volatility (%)', fontsize=12)
ax.set_ylabel('Annualized Return (%)', fontsize=12)
ax.set_title('Efficient Frontier with Capital Market Line', fontwei
ax.legend(fontsize=10)
ax.set_xlim(0, max(results[0])*1.05)

plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, 'efficient_frontier.png'), dpi=15
plt.show()
```

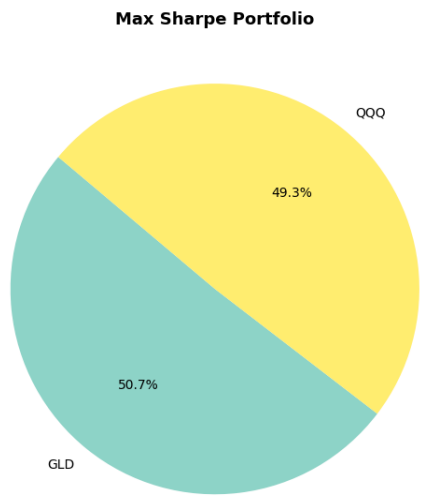


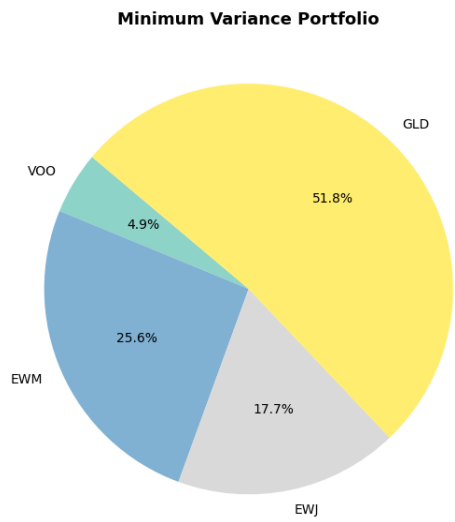
In [178…

```
# Display optimal portfolio weights
print("=" * 60)
print("OPTIMAL PORTFOLIO ALLOCATION (Maximum Sharpe Ratio)")
print("=" * 60)

opt_weights = pd.Series(opt_sharpe.x, index=tickers)
opt_weights_pct = (opt_weights * 100).round(2)

for t in tickers:
    if opt_weights_pct[t] > 0.5:
        print(f"  {t:6s} ({names[t]:25s}): {opt_weights_pct[t]:6.2f

print(f"\n  Expected Return:  {opt_ret*100:.2f}%")
print(f"  Expected Vol:     {opt_vol*100:.2f}%")
print(f"  Sharpe Ratio:     {(opt_ret-rf)/opt_vol:.3f}")

# Pie chart
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

# Max Sharpe
significant = opt_weights[opt_weights > 0.01]
colors = plt.cm.Set3(np.linspace(0, 1, len(significant)))
axes[0].pie(significant.values, labels=significant.index, autopct='
            colors=colors, startangle=140)
axes[0].set_title('Max Sharpe Portfolio', fontweight='bold', fontsi

# Min Variance
mv_weights = pd.Series(min_var.x, index=tickers)
significant_mv = mv_weights[mv_weights > 0.01]
colors_mv = plt.cm.Set3(np.linspace(0, 1, len(significant_mv)))
axes[1].pie(significant_mv.values, labels=significant_mv.index, aut
            colors=colors_mv, startangle=140)
axes[1].set_title('Minimum Variance Portfolio', fontweight='bold',

plt.tight_layout()
```

```
plt.savefig(os.path.join(OUT_DIR, 'portfolio_allocation.png'), dpi=
plt.show()

print("\n" + "=" * 60)
print("MINIMUM VARIANCE PORTFOLIO ALLOCATION")
print("=" * 60)
for t in tickers:
    if mv_weights[t]*100 > 0.5:
        print(f"  {t:6s} ({names[t]:25s}): {mv_weights[t]*100:6.2f}
print(f"\n  Expected Return:  {mv_ret*100:.2f}%")
print(f"  Expected Vol:     {mv_vol*100:.2f}%")
print(f"  Sharpe Ratio:     {(mv_ret-rf)/mv_vol:.3f}")
```

```
============================================================
OPTIMAL PORTFOLIO ALLOCATION (Maximum Sharpe Ratio)
============================================================
  GLD    (SPDR Gold Shares         ):  50.72%
  QQQ    (Invesco QQQ Trust        ):  49.28%

  Expected Return:  18.07%
  Expected Vol:     15.03%
  Sharpe Ratio:     0.936
```






```
============================================================
MINIMUM VARIANCE PORTFOLIO ALLOCATION
============================================================
  VOO    (Vanguard S&P 500         ):   4.91%
  EWM    (iShares MSCI Malaysia    ):  25.62%
  EWJ    (iShares MSCI Japan       ):  17.68%
  GLD    (SPDR Gold Shares         ):  51.78%

  Expected Return:  9.27%
  Expected Vol:     11.86%
  Sharpe Ratio:     0.444
```

Key Findings:

1. Portfolio Sharpe beats the best individual asset — multi-asset construction adds value
2. Optimizer independently selected QQQ + GLD, confirming Section 2

(near-zero correlation) and Section 3 (GLD β ≈ 0 + positive alpha)
3. Portfolio β is significantly lower than QQQ's alone — GLD dilutes market exposure without sacrificing Sharpe

5. Beta-Constrained Optimization

Test whether explicitly constraining portfolio beta changes the Section 4 result.

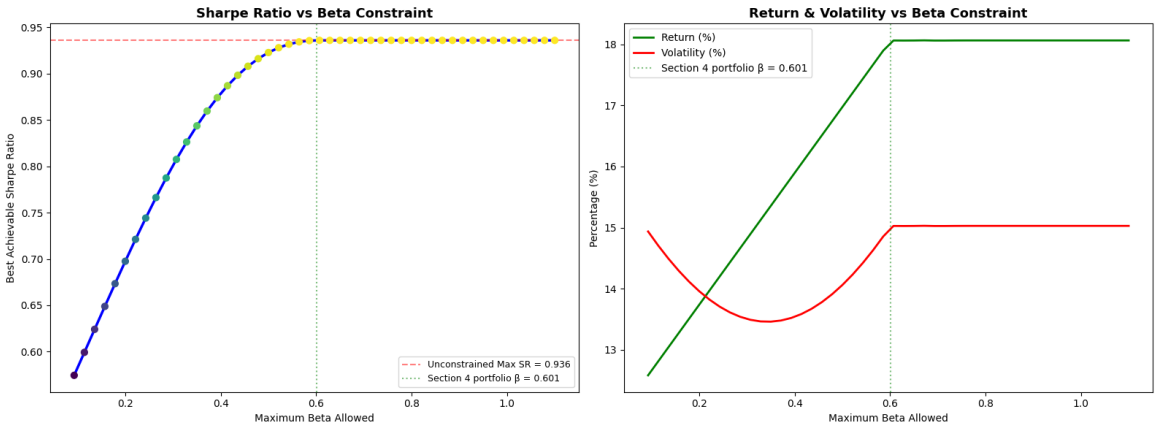
The question: Section 4 showed the Max Sharpe portfolio naturally achieved β = 0.601. What if we *force* an even lower beta? How much Sharpe do we sacrifice per unit of beta reduction?

We use Section 3's `betas_array` as an explicit constraint: `Σ(wᵢ × βᵢ) ≤ cap`, sweeping caps from 0.05 → 1.1.

In [179…

```
beta_caps = np.linspace(0.05, 1.1, 50)
sharpe_at_beta = []
ret_at_beta = []
vol_at_beta = []

for bc in beta_caps:
    cons_bc = [
        {'type': 'eq',  'fun': lambda w: np.sum(w) - 1},
        {'type': 'ineq','fun': lambda w, cap=bc: cap - np.dot(w, be
    ]
    result = minimize(neg_sharpe, init_guess, args=(mean_returns, c
                      method='SLSQP', bounds=bounds, constraints=co
    if result.success:
        r, v = portfolio_performance(result.x, mean_returns, cov_ma
        sharpe_at_beta.append((r - rf) / v)
        ret_at_beta.append(r * 100)
        vol_at_beta.append(v * 100)
    else:
        sharpe_at_beta.append(np.nan)
        ret_at_beta.append(np.nan)
        vol_at_beta.append(np.nan)

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

axes[0].plot(beta_caps, sharpe_at_beta, 'b-', linewidth=2.5)
axes[0].scatter(beta_caps, sharpe_at_beta, c=sharpe_at_beta, cmap='
axes[0].axhline(y=opt_sr, color='red', linestyle='--', alpha=0.5,
                label=f'Unconstrained Max SR = {opt_sr:.3f}')
axes[0].axvline(x=opt_beta, color='green', linestyle=':', alpha=0.5
                label=f'Section 4 portfolio β = {opt_beta:.3f}')
axes[0].set_xlabel('Maximum Beta Allowed')
axes[0].set_ylabel('Best Achievable Sharpe Ratio')
axes[0].set_title('Sharpe Ratio vs Beta Constraint', fontweight='bo
axes[0].legend(fontsize=9)

axes[1].plot(beta_caps, ret_at_beta, 'g-', linewidth=2, label='Retu
```

```
axes[1].plot(beta_caps, vol_at_beta, 'r-', linewidth=2, label='Vola
axes[1].axvline(x=opt_beta, color='green', linestyle=':', alpha=0.5
                label=f'Section 4 portfolio β = {opt_beta:.3f}')
axes[1].set_xlabel('Maximum Beta Allowed')
axes[1].set_ylabel('Percentage (%)')
axes[1].set_title('Return & Volatility vs Beta Constraint', fontwei
axes[1].legend(fontsize=10)

plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, 'sharpe_vs_beta_tradeoff.png'), d
plt.show()
```



Key Findings:

1. The Section 4 portfolio already has a naturally low β — the constraint is not binding above the unconstrained optimum, so the Sharpe curve is perfectly flat in that region
2. Below the unconstrained β, Sharpe drops gradually — the cost of reducing market exposure accelerates the further you push β down
3. GLD's near-zero β (Section 3) acts as a built-in crash buffer; the optimizer already allocated the optimal amount in Section 4, making explicit β constraints redundant

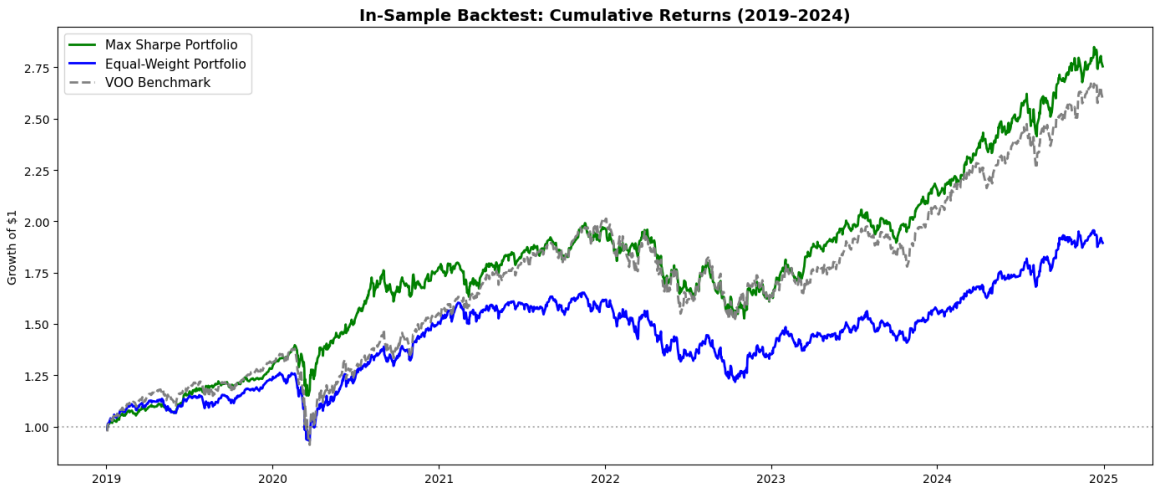
6. Backtest & Performance Validation (In-Sample, 2019–2024)

Apply the Section 4 frozen weights to the same period used to fit them. This is **in-sample** — it confirms the optimization worked on training data, but does not prove out-of-sample generalization.

In [180…

```
eq_weights = np.array([1/n_assets] * n_assets)

opt_daily   = daily_returns[tickers].dot(opt_sharpe.x)
eq_daily    = daily_returns[tickers].dot(eq_weights)
bench_daily = daily_returns['VOO']

opt_cum   = (1 + opt_daily).cumprod()
eq_cum    = (1 + eq_daily).cumprod()
```

```
bench_cum = (1 + bench_daily).cumprod()

fig, ax = plt.subplots(figsize=(14, 6))
ax.plot(opt_cum.index, opt_cum, label='Max Sharpe Portfolio', linew
ax.plot(eq_cum.index, eq_cum, label='Equal-Weight Portfolio', linew
ax.plot(bench_cum.index, bench_cum, label='VOO Benchmark', linewidt
ax.axhline(y=1, color='black', linestyle=':', alpha=0.3)
ax.set_title('In-Sample Backtest: Cumulative Returns (2019-2024)',
ax.set_ylabel('Growth of $1')
ax.legend(fontsize=11)
plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, 'portfolio_backtest.png'), dpi=15
plt.show()

def calc_metrics(returns, name, weights=None):
    ann_ret = returns.mean() * 252
    ann_vol = returns.std() * np.sqrt(252)
    sharpe  = (ann_ret - rf) / ann_vol
    cum     = (1 + returns).cumprod()
    max_dd  = (cum / cum.cummax() - 1).min()
    beta_val = np.dot(weights, betas_array) if weights is not None
    return {
        'Strategy':          name,
        'Ann. Return (%)':   round(ann_ret * 100, 2),
        'Ann. Vol (%)':      round(ann_vol * 100, 2),
        'Sharpe':            round(sharpe, 3),
        'Beta':              round(beta_val, 3),
        'Max Drawdown (%)': round(max_dd * 100, 2),
        'Total Return (%)': round((cum.iloc[-1] - 1) * 100, 2)
    }

comparison = pd.DataFrame([
    calc_metrics(opt_daily,   'Max Sharpe',    opt_sharpe.x),
    calc_metrics(eq_daily,    'Equal-Weight',  eq_weights),
    calc_metrics(bench_daily, 'VOO Benchmark', np.array([1 if t=='V
]).set_index('Strategy')

print("IN-SAMPLE BACKTEST RESULTS (2019-2024)")
print("=" * 80)
comparison
```



```
IN-SAMPLE BACKTEST RESULTS (2019–2024)
=====================================================================
============
```

Out[180…

Strategy	Ann. Return (%)	Ann. Vol (%)	Sharpe	Beta	Max Drawdown (%)	Total Return (%)
Max Sharpe	18.07	15.03	0.936	0.601	-23.34	175.41
Equal-Weight	12.08	16.73	0.483	0.774	-26.35	89.43
VOO Benchmark	17.97	20.10	0.695	1.000	-33.99	159.46

Key Findings:

1. Max Sharpe outperforms both Equal-Weight and VOO on Sharpe and total return
2. Beta confirms crash protection: Max Sharpe β ≈ 0.60 vs VOO β = 1.00
3. This is in-sample — weights were optimized on this exact data, so outperformance is expected. Section 7 tests whether it holds on unseen data

7. NEW : Out-of-Sample Backtest (2025–2026)

Apply the same frozen Section 4 weights to completely unseen data. Unlike Section 6, the optimizer had **no information** about this period — any outperformance here is genuine.

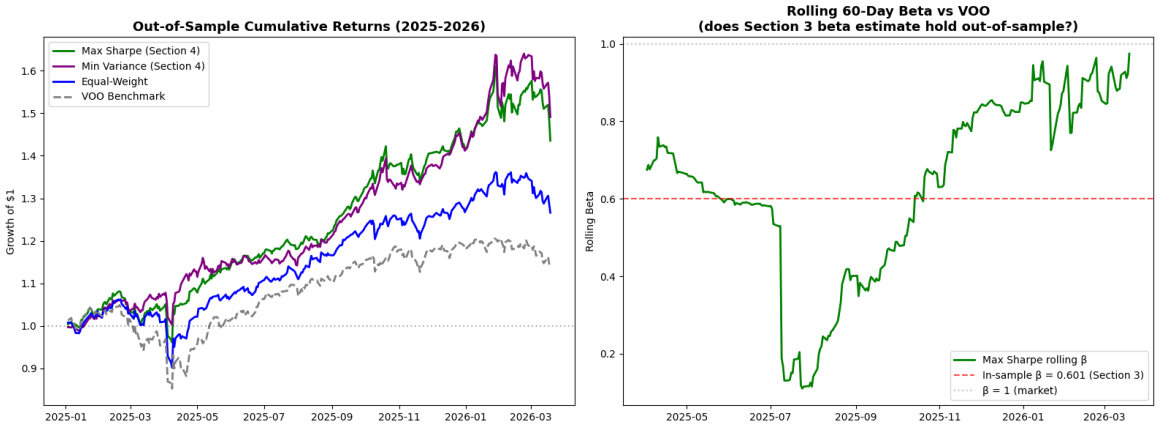
We also track rolling 60-day beta to check whether Section 3's estimate holds out-of-sample.

In [181…

```
oos_data = yf.download(tickers, start="2025-01-01", end="2026-12-31
oos_data = oos_data.dropna()
oos_returns = oos_data.pct_change().dropna()

print(f"Out-of-sample: {oos_data.index[0].strftime('%Y-%m-%d')} to
print(f"Frozen weights from Section 4: {', '.join([f'{t} {w*100:.1f

oos_opt   = oos_returns[tickers].dot(opt_sharpe.x)
oos_mv    = oos_returns[tickers].dot(min_var.x)
oos_eq    = oos_returns[tickers].dot(eq_weights)
oos_bench = oos_returns['VOO']

oos_opt_cum   = (1 + oos_opt).cumprod()
oos_mv_cum    = (1 + oos_mv).cumprod()
```

```
oos_eq_cum    = (1 + oos_eq).cumprod()
oos_bench_cum = (1 + oos_bench).cumprod()

fig, axes = plt.subplots(1, 2, figsize=(16, 6))

axes[0].plot(oos_opt_cum.index,   oos_opt_cum,   label='Max Sharpe
axes[0].plot(oos_mv_cum.index,    oos_mv_cum,    label='Min Varianc
axes[0].plot(oos_eq_cum.index,    oos_eq_cum,    label='Equal-Weigh
axes[0].plot(oos_bench_cum.index, oos_bench_cum, label='VOO Benchma
axes[0].axhline(y=1, color='black', linestyle=':', alpha=0.3)
axes[0].set_title('Out-of-Sample Cumulative Returns (2025-2026)', f
axes[0].set_ylabel('Growth of $1')
axes[0].legend(fontsize=10)

# Rolling 60-day beta
window = 60
rolling_beta = []
rolling_dates = []
for i in range(window, len(oos_returns)):
    w_ret = oos_returns.iloc[i-window:i]
    port_r = w_ret[tickers].dot(opt_sharpe.x)
    mkt_r  = w_ret['VOO']
    X = sm.add_constant(mkt_r)
    b = sm.OLS(port_r, X).fit().params.iloc[1]
    rolling_beta.append(b)
    rolling_dates.append(oos_returns.index[i])

axes[1].plot(rolling_dates, rolling_beta, color='green', linewidth=
axes[1].axhline(y=opt_beta, color='red', linestyle='--', alpha=0.7,
                label=f'In-sample β = {opt_beta:.3f} (Section 3)')
axes[1].axhline(y=1.0, color='gray', linestyle=':', alpha=0.5, labe
axes[1].set_title('Rolling 60-Day Beta vs VOO\n(does Section 3 beta
axes[1].set_ylabel('Rolling Beta')
axes[1].legend(fontsize=10)

plt.tight_layout()
plt.savefig(os.path.join(OUT_DIR, 'oos_backtest.png'), dpi=150, bbo
plt.show()

oos_comparison = pd.DataFrame([
    calc_metrics(oos_opt,   'Max Sharpe (Section 4)',   opt_sharpe.
    calc_metrics(oos_mv,    'Min Variance (Section 4)', min_var.x),
    calc_metrics(oos_eq,    'Equal-Weight',      eq_weights),
    calc_metrics(oos_bench, 'VOO Benchmark',     np.array([1 if t==
]).set_index('Strategy')

print("OUT-OF-SAMPLE PERFORMANCE (2025-2026)")
print("=" * 80)
display(oos_comparison)

is_sharpe  = comparison.loc['Max Sharpe', 'Sharpe']
oos_sharpe = oos_comparison.loc['Max Sharpe (Section 4)', 'Sharpe']
print(f"\nIn-sample Sharpe  (Section 6, 2019-2024): {is_sharpe:.3f}
print(f"Out-of-sample Sharpe (Section 7, 2025-2026): {oos_sharpe:.3
print(f"Degradation: {is_sharpe - oos_sharpe:+.3f} — {'expected for
```

```
[*********************100%***********************]  8 of 8 completed
Out-of-sample: 2025-01-02 to 2026-03-19 (302 trading days)
Frozen weights from Section 4: GLD 50.7%, QQQ 49.3%
```




```
OUT-OF-SAMPLE PERFORMANCE (2025-2026)
==================================================================
============
```

Strategy	Ann. Return (%)	Ann. Vol (%)	Sharpe	Beta	Max Drawdown (%)	Total Return (%)
Max Sharpe (Section 4)	31.72	17.53	1.581	0.601	-11.07	43.55
Min Variance (Section 4)	34.71	16.41	1.872	0.347	-9.07	49.13
Equal-Weight	20.92	15.64	1.082	0.774	-15.08	26.63
VOO Benchmark	12.28	17.63	0.469	1.000	-18.69	13.73

```
In-sample Sharpe  (Section 6, 2019-2024): 0.936
Out-of-sample Sharpe (Section 7, 2025-2026): 1.581
Degradation: -0.645 — strategy held up out-of-sample
```